# Differential Equations

Differential equation - An equation that contains an unknown function and some of its derivatives

Ex: Population growth can be modeled with this differential equation

$\frac{dP}{dt} = kP$ where $k$ is the relative growth rate.

What function whos derivative is a constant multiple of itself?

$P(t) = Ce^{kt}$ $\quad P'(t) = k(Ce^{kt}) = kP$

$P(0) = Ce^{k0} = C$ so $C$ is the starting population.

But populations usually level off once they reach their carrying capacity ($M$)

Logistic differential equation: $\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$

$\frac{dP}{dt} \approx kP$ when $P$ is small $\quad \frac{dP}{dt} < 0$ when $P > M$

Logistic growth equation: $P(t) = \frac{M}{1 + Ae^{-kt}}$ where $A = \frac{M - P(0)}{P(0)}$

Ex 2: The motion of a spring

Force of spring $= -kx = ma = m\frac{d^2x}{dt^2}$

$\frac{d^2x}{dt^2} = -\frac{k}{m}x$ what function whos 2nd derivative is a constant multiplied by the negative of itself?

Sin or cos

## Direction/Slope Fields

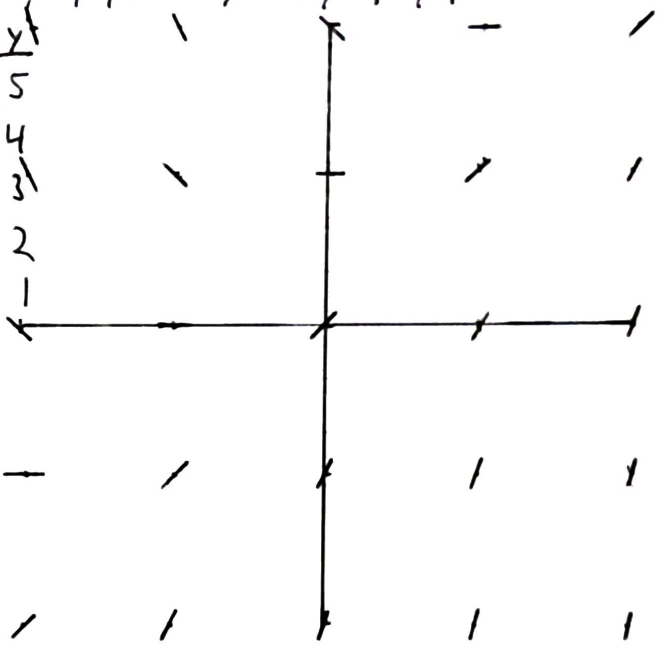
Drawing short line segments with the slope at many points.

Ex: $y' = x - y + 1$ $\quad x = -2, -1, 0, 1, 2 \quad y = -2, -1, 0, 1, 2$

| x | y | y' |
|---|---|---|
| -2 | -2 | 1 |
| -2 | -1 | 0 |
| -2 | 0 | -1 |
| -2 | 1 | -2 |
| -2 | 2 | -3 |
| -1 | -2 | 2 |
| -1 | -1 | 1 |
| -1 | 0 | 0 |
| -1 | 1 | -1 |
| -1 | 2 | -2 |

| x | y | y' |
|---|---|---|
| 0 | -2 | 3 |
| 0 | -1 | 2 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 0 | 2 | -1 |
| 1 | -2 | 4 |
| 1 | -1 | 3 |
| 1 | 0 | 2 |
| 1 | 1 | 1 |
| 1 | 2 | 0 |

| x | y | y' |
|---|---|---|
| 2 | -2 | 5 |
| 2 | -1 | 4 |
| 2 | 0 | 3 |
| 2 | 1 | 2 |
| 2 | 2 | 1 |



## Euler's Method

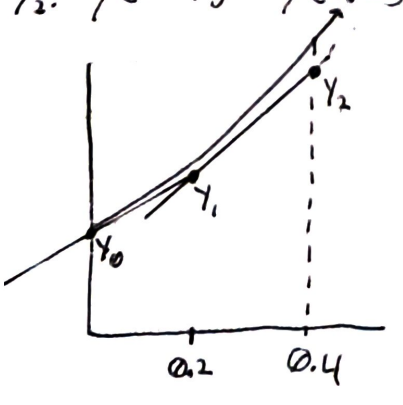
Approximating a value of the function from an initial value.

$$y_1 = y_0 + \Delta x\, f'(x_0, y_0)$$

Ex: $y' = y \quad y(0) = 1 \quad y(0.4) = ? \quad \Delta x = 0.2$

$y_1$: $y(0.2) = y(0) + \Delta x\, y'(0, 1) = 1 + 0.2(1) = 1.2$

$y_2$: $y(0.4) = y(0.2) + \Delta x\, y'(0.2, 1.2) = 1.2 + 0.2(1.2) = 1.44$




$y = e^x \qquad e^{0.4} \approx 1.44$

The more steps, the further the error